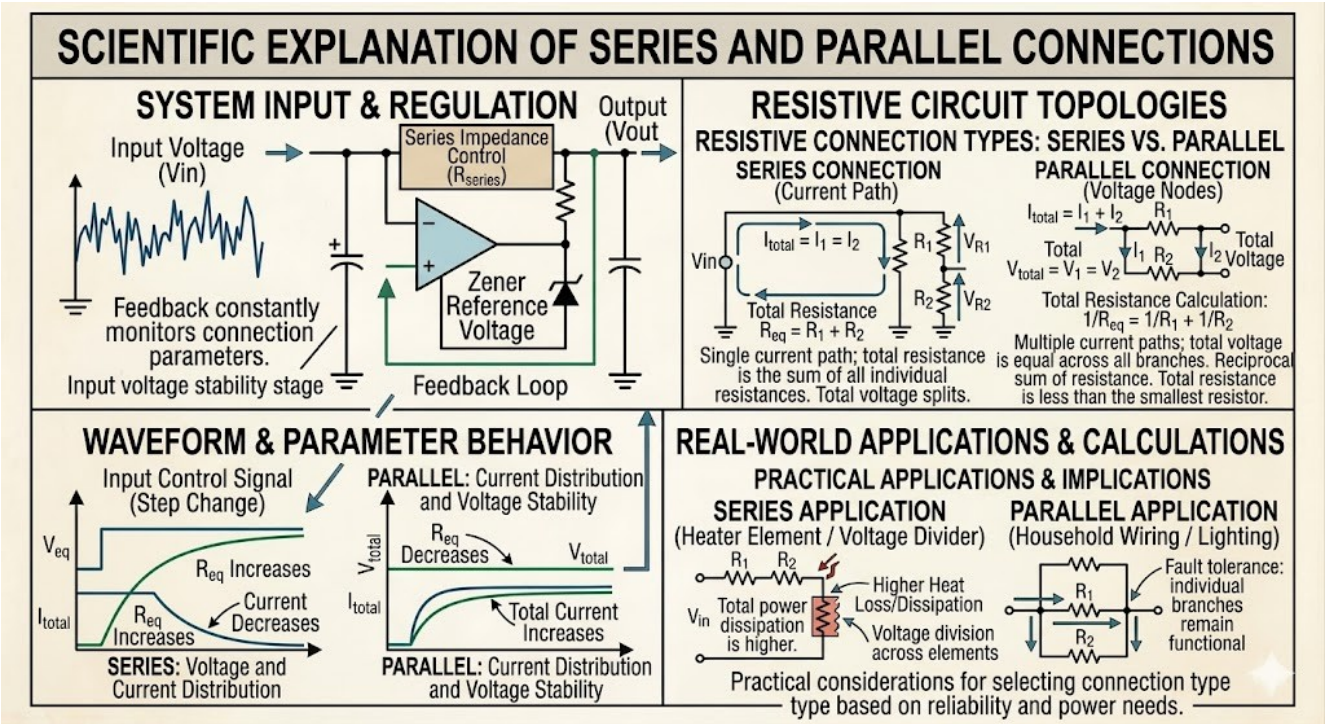


Physics Guide: COMPREHENSIVE TECHNICAL REPORT:SERIES AND PARALEL ANALOG

Classification: TECHNICAL GUIDE/ EDUCATIONAL

Xumar Bağıyeva, Aysel İsmayılova, Rəşadə Zülfüqarlı, Yasin Quluzade,Ahmed Abdullayev

Verified by: Physics Teacher Azerbaijan Telman Askeraliyev (Fizika Muellimi, Azerbaijan, Baku)



In electrical circuit design, components can be arranged in two fundamental topological configurations: **Series** and **Parallel**. The chosen network configuration directly dictates how equivalent resistance, system current, and voltage drops are distributed across the network.

- **Input Voltage (V_{in}) & Input Stage:** The raw electrical energy enters the network structure. Parallel filtering capacitors are implemented at the input boundary to stabilize the voltage rail and suppress transient noise before the signal enters the main resistive topologies.

- **Series Impedance Control (R_{series}):** In linear regulation configurations, a series network is often monitored by a differential operational amplifier. The op-amp evaluates the load parameters against a highly stable **Zener Reference Voltage**.
 - **Closed Feedback Loop:** A dedicated feedback loop continuously tracks the connection parameters. If the configuration's dynamic load shifts, the loop triggers corrective resistance modulation via the control amplifier, ensuring a stable output voltage (V_{out}).
-

III SECTION 2: RESISTIVE CIRCUIT TOPOLOGIES

The behavior of electrical circuits changes drastically depending on how the conductive paths are physically routed:

1. Series Connection (Single Current Path)

- **Structure:** Components are connected end-to-end in a sequential chain, forming a single, continuous loop for electrical charge carriers.
- **Operational Characteristics:** Because there is only one path, the total current (I_{total}) passing through every single component is identical ($I_{\text{total}} = I_1 = I_2$).
- **Voltage and Resistance:** The total input voltage is divided among the components based on their individual values ($V_{\text{total}} = V_{R1} + V_{R2}$). The total equivalent resistance (R_{eq}) is simply the direct sum of all individual resistances ($R_{\text{eq}} = R_1 + R_2$), meaning adding more components increases total loop impedance.

2. Parallel Connection (Independent Voltage Nodes)

- **Structure:** Components are connected side-by-side across the same two electrical nodes, creating multiple branching paths.
- **Operational Characteristics:** The voltage potential across every single parallel branch is identical to the total source voltage ($V_{\text{total}} = V_1 = V_2$).
- **Current and Resistance:** The total system current splits among the available branches based on their relative conductivity ($I_{\text{total}} = I_1 + I_2$). The equivalent resistance drops as more branches are added.

because more paths are opened for current. It is calculated using the reciprocal formula:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

(Consequently, the total equivalent resistance of a parallel network is always strictly less than the resistance of its smallest individual branch).

📖 SECTION 3: WAVEFORM & PARAMETER BEHAVIOR

When a step-change or an input control signal is introduced into the circuit, the electrical parameter distribution reacts dynamically according to the connection type:

1. Series Network Response

- **Behavior:** When more resistive elements are added in series, the system's equivalent resistance (R_{eq}) steps up instantly.
- **Impact:** In accordance with Ohm's Law, as total resistance increases under a constant voltage, the total network current (I_{total}) experiences a sharp, corresponding decrease.

2. Parallel Network Response

- **Behavior:** When additional load branches are activated in a parallel configuration, the overall equivalent resistance (R_{eq}) decreases.
 - **Impact:** Because the voltage across the nodes remains perfectly stable, opening new conduction channels causes the total supply current drawn by the system (I_{total}) to increase proportionally.
-

✂ SECTION 4: REAL-WORLD APPLICATIONS & IMPLEMENTATIONS

1. Series Applications (Voltage Division & Safety)

- **Heater Elements & Industrial Heating:** Multiple heating elements are wired in series to evenly distribute high voltages and safely share power dissipation.
- **Voltage Dividers:** Utilized extensively in sensor signal conditioning (such as LDR light-sensing circuits) to scale down high analog voltage levels into readable microcontroller inputs.

- **Critical Flaw:** If a single component fails or opens in a series loop, the entire current path is broken, causing all subsequent components to shut down instantly.

2. Parallel Applications (Household Grids & Fault Tolerance)

- **Household Wiring and Lighting Grid:** All residential appliances and light fixtures are connected in parallel. This guarantees that every device receives the full rated mains voltage (220V or 110V) regardless of how many appliances are active.
- **Fault Tolerance:** If a single branch or lightbulb burns out in a parallel circuit, the remaining branches maintain closed connections to the main voltage nodes and continue to operate completely unaffected.